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# 1 Objectives

1. **Determination of Buckling Loads:**
   - Investigate and quantify the buckling loads of slender pin-ended bars through experimental procedures.
   - Analyze the relationship between applied loads and the resulting center deflections for each tested bar.
2. **Comparison of Theoretical and Experimental Critical Loads:**
   - Assess and calculate the theoretical critical loads for the pin-ended bars using relevant analytical models.
   - Perform experimental tests to determine the critical loads for each bar.
   - Conduct a comprehensive comparison between the theoretical predictions and the experimentally obtained critical loads for all tested bars.

Through these objectives, the aim is to enhance our understanding of the buckling behavior of slender pin-ended bars and to evaluate the accuracy of theoretical predictions in relation to practical experimental outcomes.

# 2 Procedure

1. **Bar Insertion and Setup:**
   - Insert one of the bars between the supports, ensuring that the ball bearing at the ends seats firmly.
   - Orient the larger dimension of the cross-section parallel to the floor.
   - Position the displacement transducer to measure the vertical deflection of the bar.
2. **Moment of Inertia Calculation and Critical Load Prediction:**
   - Calculate the minimum moment of inertia for the inserted bar.
   - Predict the critical load for the bar based on the calculated moment of inertia.
3. **Data Collection Using Dalite Datascan Configurator:**
   - Open the Dalite Datascan Configurator at the DB2Bc.OVL file.
   - Click on the "monitor every second" window.
   - Verify that the initial reading for the electrical potentiometer is zero.
   - Apply the load increments slowly.
   - At each increment, record the central deflection.
   - Continue to increase the load until the column buckles or the central deflection becomes large.
4. **Repeat for All Bars:**
   - Repeat steps 2 to 4 for all four bars in the experimental setup.

This procedure outlines the steps to insert the bars, calculate relevant parameters, and conduct the experimental testing using the Dalite Datascan Configurator. It ensures systematic data collection for each bar in the study.

# 3 Results

## 3.1 Plot Calibration Charts

The calibration charts for the load cell and the displacement transducer are shown in the subsequent figures. The calibration charts are used to determine the equation of the line of best fit for the data points. The equation is then used to calculate the load for any pressure values that exceeded those for which the corresponding load is provided.

### 3.1.1 Bar 1: 508 mm

The calibration chart for the load cell of Bar 1 with 508 mm length is shown in figure 1 along with the equation of the line of best fit.

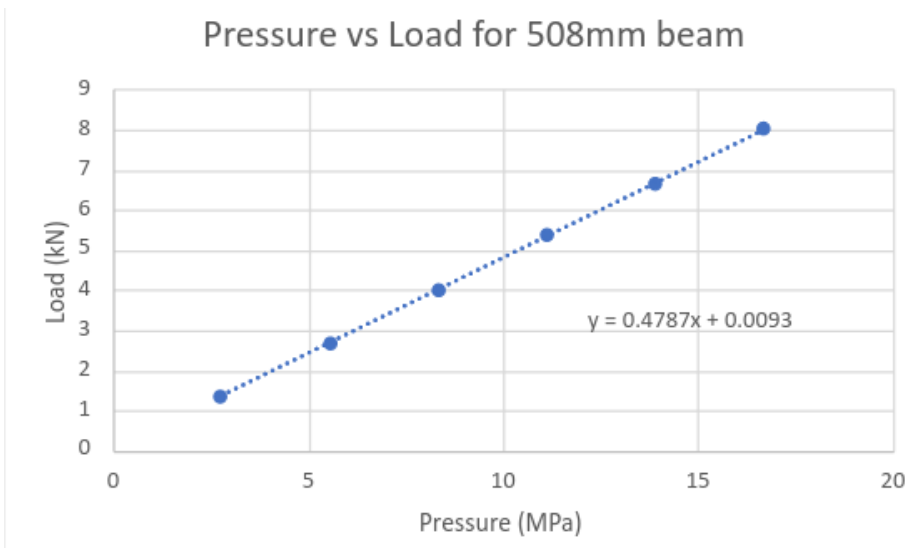


Figure 1: Bar 1 Calibration Chart

### 3.1.2 Bar 2: 737 mm

The calibration chart for the load cell of Bar 2 with 737 mm length is shown in figure 2 along with the equation of the line of best fit.

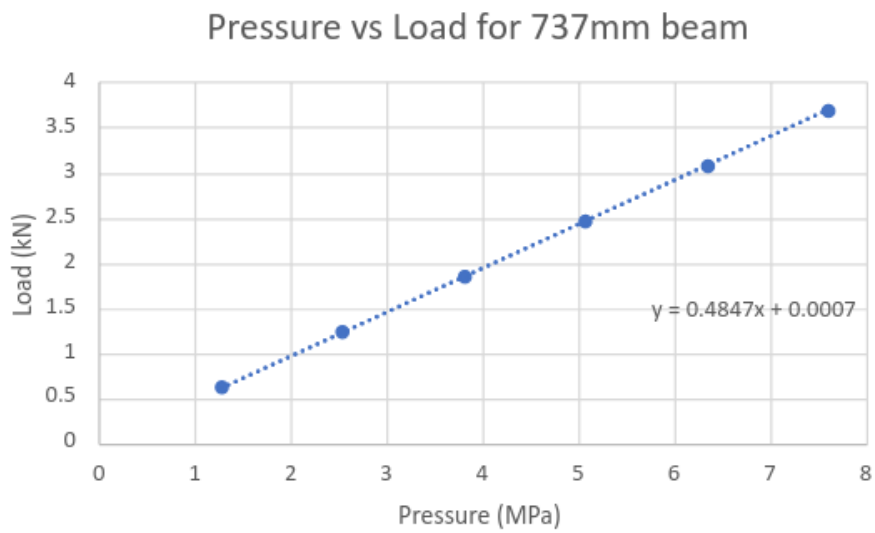


Figure 2: Bar 2 Calibration Chart

### 3.1.3 Bar 3: 902 mm

The calibration chart for the load cell of Bar 3 with 902 mm length is shown in figure 3 along with the equation of the line of best fit.

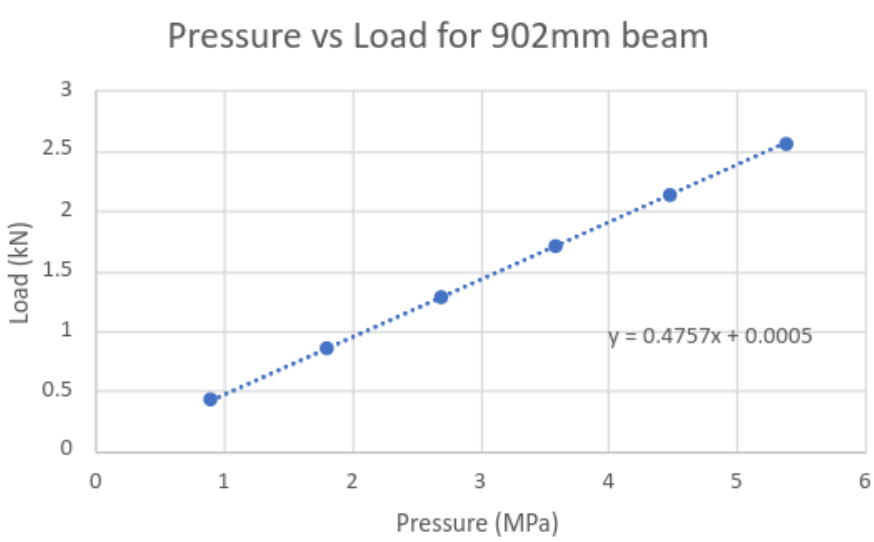


Figure 3: Bar 3 Calibration Chart

### 3.1.4 Bar 4: 997 mm

The calibration chart for the load cell of Bar 4 with 997 mm length is shown in figure 4 along with the equation of the line of best fit.

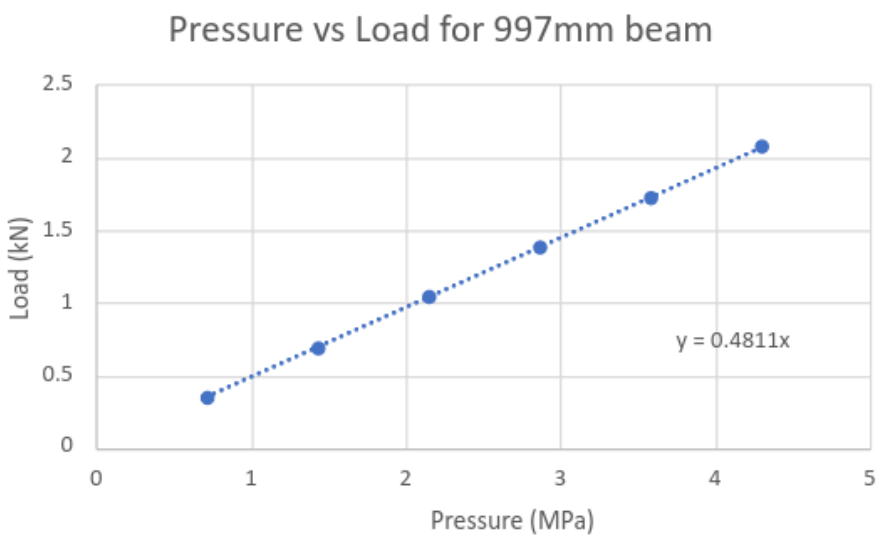


Figure 4: Bar 4 Calibration Chart

## 3.2 Calculated Pressure for Exceeding Loads

The pressure is calculated using the equation of the line of best fit for the calibration charts. The calculated pressure is shown along with the load and the deflection for each exceeding load in the subsequent tables.

### 3.2.1 Bar 1: 508 mm

The calculated pressure for the exceeding loads of Bar 1 with 508 mm length is shown in table 1.

Table 1: Bar 1: 508 mm Exceeding Loads

| **Pressure (MPa)** | **Load (kN)** | **Deflection (mm)** |
|---|---|---|
| 2.785481949 | 1.34 | 0.3277 |
| 5.570963899 | 2.68 | 0.7734 |
| 8.356445848 | 4.015 | 1.271 |
| 11.1419278 | 5.35 | 1.9436 |
| 13.92740975 | 6.64 | 2.7734 |
| 16.7128917 | 8.03 | 4.7394 |
| 17.71952626 | 8.491637222 | 12.083 |

### 3.2.2 Bar 2: 737 mm

The calculated pressure for the exceeding loads of Bar 2 with 737 mm length is shown in table 2.

Table 2: Bar 2: 737 mm Exceeding Loads

| **Pressure (MPa)** | **Load (kN)** | **Deflection (mm)** |
|---|---|---|
| 1.268635343 | 0.616 | 0.2422 |
| 2.537270687 | 1.23 | 0.6727 |
| 3.80590603 | 1.845 | 1.2167 |
| 5.074541373 | 2.46 | 1.6104 |
| 6.343176716 | 3.075 | 2.1479 |
| 7.61181206 | 3.69 | 2.7867 |
| 10.06634566 | 4.879857741 | 23.3818 |

### 3.2.3 Bar 3: 902 mm

The calculated pressure for the exceeding loads of Bar 3 with 902 mm length is shown in table 3.

Table 3: Bar 3: 737 mm Exceeding Loads

| **Pressure (MPa)** | **Load (kN)** | **Deflection (mm)** |
|---|---|---|
| 0.896318449 | 0.427 | 0.1628 |
| 1.792636898 | 0.853 | 0.2696 |
| 2.688955347 | 1.28 | 0.4546 |
| 3.585273796 | 1.707 | 0.6953 |
| 4.481592245 | 2.13 | 1.2315 |
| 5.377910694 | 2.56 | 5.1589 |
| 7.322232253 | 3.483685883 | 18.8641 |

### 3.2.4 Bar 4: 997 mm

The calculated pressure for the exceeding loads of Bar 4 with 997 mm length is shown in table 4.

Table 4: Bar 4: 997 mm Exceeding Loads

| **Pressure (MPa)** | **Load (kN)** | **Deflection (mm)** |
|---|---|---|
| 0.717054759 | 0.345 | 0.1956 |
| 1.434109519 | 0.69 | 0.6738 |
| 2.151164278 | 1.035 | 1.0114 |
| 2.868219037 | 1.38 | 1.3878 |
| 3.585273796 | 1.725 | 1.9515 |
| 4.302328556 | 2.07 | 2.4975 |
| 6.343176716 | 3.051702318 | 20.7397 |

## 3.3 Calculate Minimum Moment of Inertia

## 3.4 Plot Load-Deflection Curves

This section shows the load-deflection curves for each bar with an indication of the critical load. Then, the percentage error between the theoretical and experimental critical loads is calculated.

### 3.4.1 Bar 1: 508 mm

The load-deflection curve for Bar 1 with 508 mm length is shown in figure 5. The theoretical critical load is 8.4916 kN and the experimental critical load is 8.03 kN.

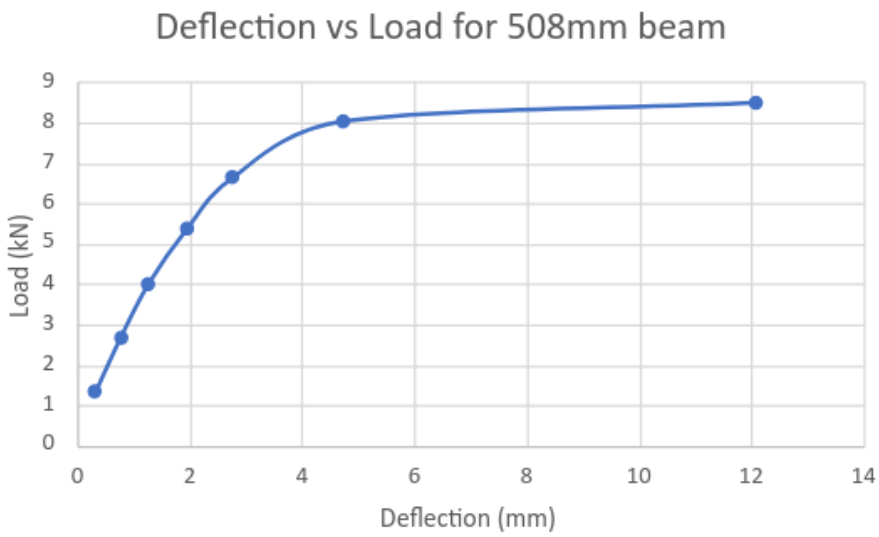


Figure 5: Bar 1: 508 mm Load-Deflection Curve

The percentage error between the theoretical and experimental critical loads is calculated using equation 1 and is shown in equation 2.

$$\text{Percentage Error} = \frac{\text{Experimental} - \text{Theoretical}}{\text{Theoretical}} \times 100 \tag{1}$$

Substituting the values in equation 1 yields equation 2.

$$\text{Percentage Error} = \frac{|8.03 - 8.4916|}{|8.4916|} \times 100 = 5.43\% \tag{2}$$

### 3.4.2 Bar 2: 737 mm

The load-deflection curve for Bar 2 with 737 mm length is shown in figure 6. The theoretical critical load is 4.8799 kN and the experimental critical load is 3.69 kN.

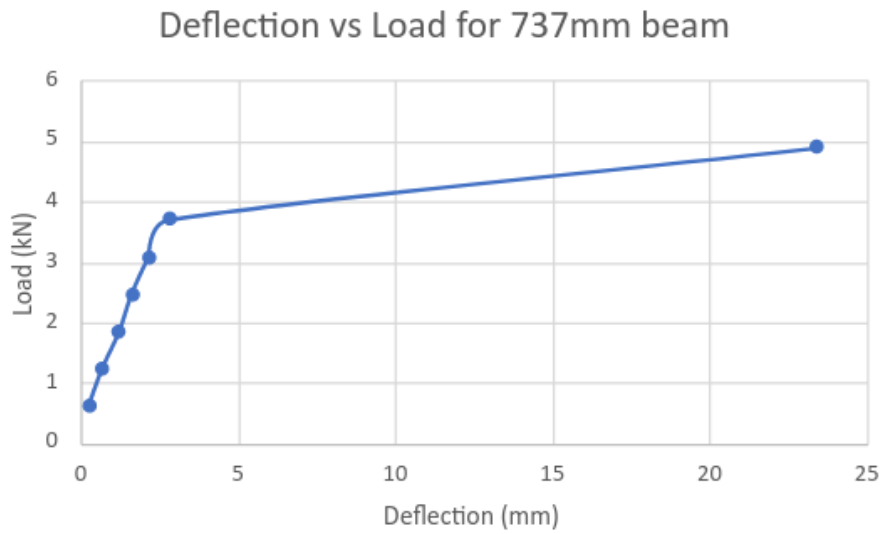


Figure 6: Bar 2: 737 mm Load-Deflection Curve

The percentage error between the theoretical and experimental critical loads is calculated using equation 1 and is shown in equation 3.

$$\text{Percentage Error} = \frac{|3.69 - 4.8799|}{|4.8799|} \times 100 = 24.38\% \tag{3}$$

### 3.4.3 Bar 3: 902 mm

The load-deflection curve for Bar 3 with 902 mm length is shown in figure 7. The theoretical critical load is 3.4837 kN and the experimental critical load is 2.56 kN.

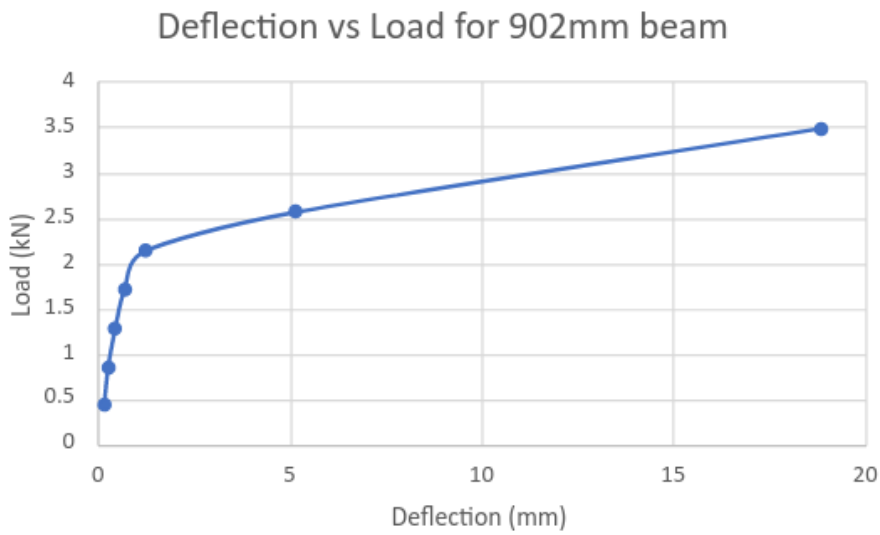


Figure 7: Bar 3: 902 mm Load-Deflection Curve

The percentage error between the theoretical and experimental critical loads is calculated using equation 1 and is shown in equation 4.

$$\text{Percentage Error} = \frac{|2.56 - 3.4837|}{|3.4837|} \times 100 = 26.51\% \tag{4}$$

### 3.4.4 Bar 4: 997 mm

The load-deflection curve for Bar 4 with 997 mm length is shown in figure 8. The theoretical critical load is 3.0517 kN and the experimental critical load is 2.07 kN.

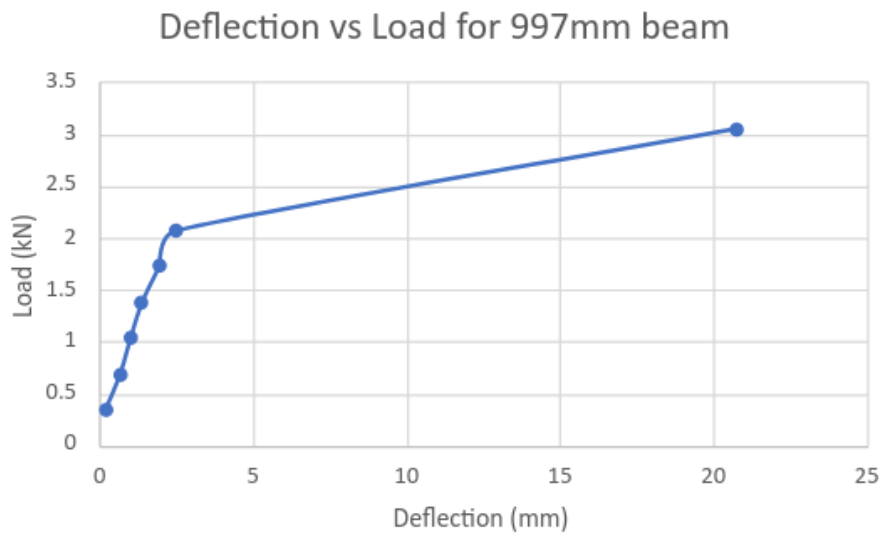


Figure 8: Bar 4: 997 mm Load-Deflection Curve

The percentage error between the theoretical and experimental critical loads is calculated using equation 1 and is shown in equation 5.

$$\text{Percentage Error} = \frac{|2.07 - 3.0517|}{|3.0517|} \times 100 = 32.13\% \tag{5}$$

## 3.5 Plot Critical Stress-Slenderness Ratio Curves

This section shows the critical stress-slenderness ratio curves for each bar for both the theoretical and experimental values in a single plot. The plot is shown in figure 9.

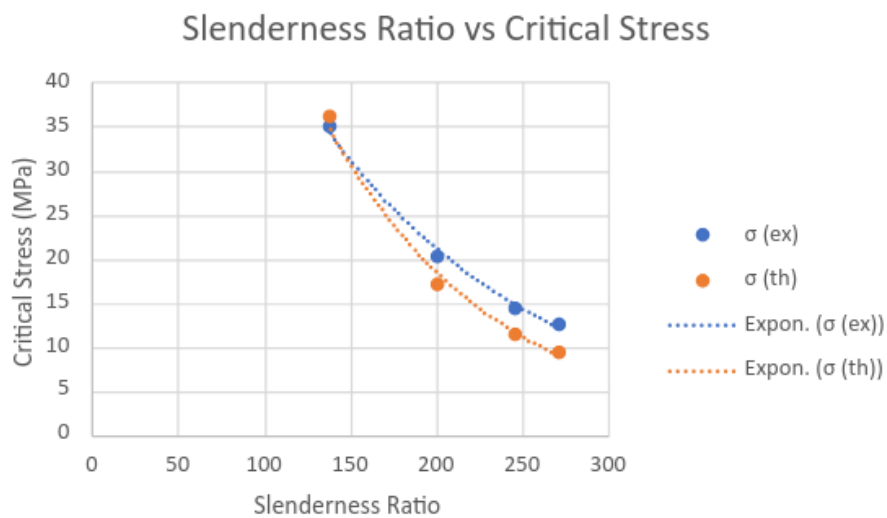


Figure 9: Critical Stress-Slenderness Ratio Curves

## 3.6 Summary of Results

In this section, the pertinent results of the experiment are summarized in a table. The table includes ($L$, $\frac{L}{r}$, $P_{cr,theo}$, $P_{cr,exp}$, $\sigma_{cr,theo}$, $\sigma_{cr,exp}$, and % Error). The summary table is shown in table 5.

Table 5: Summary of Results

| **Bar** | **L (mm)** | $\frac{L}{r}$ | $P_{cr,theo}$ **(kN)** | $P_{cr,exp}$ **(kN)** | $\sigma_{cr,theo}$ **(MPa)** | $\sigma_{cr,exp}$ **(MPa)** | **% Error** |
|---|---|---|---|---|---|---|---|
| 1 | 0.508 | 138.5640646 | 8.491637222 | 8.714670491 | 35.06202685 | 35.98293271 | 2.559285163 |
| 2 | 0.737 | 201.0269992 | 4.879857741 | 4.140410674 | 20.14896523 | 17.09578335 | 17.85926868 |
| 3 | 0.902 | 246.0330438 | 3.483685883 | 2.764173634 | 14.3841623 | 11.41329141 | 26.02992228 |
| 4 | 0.997 | 271.945615 | 3.051702318 | 2.262497347 | 12.60049927 | 9.341866671 | 34.88202858 |

# 4 Discussion

# 5 Conclusions